

Research on Ultra-Short-Term Wind Power Generation Forecasting Based on Expert System Method and RBF Neural Network Prediction Model

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Abstract—To achieve the "dual carbon" goals, China is actively promoting the use of clean energy, among which wind power generation has been rapidly applied due to its unique advantages. However, wind power generation is greatly affected by weather conditions, posing challenges to the stable power supply of the power grid. Therefore, accurate forecasting of wind power generation is extremely important. This paper uses an expert system method combined with an RBF neural network model to construct an optimized composite forecasting model to improve forecasting accuracy. Through this model, wind power generation can be accurately predicted, thus ensuring the stable operation of the power grid and improving the utilization rate of wind energy resources. The case study results show that the forecasting model is significantly effective in predicting wind power generation and helps promote the transformation to green energy and the stable operation of the power system.

Keywords - Expert System Method, RBF Neural Network, Wind Power Generation Forecasting, Combining Model

I. INTRODUCTION

In order to achieve the "dual carbon" goals, that is, peak carbon and carbon neutrality, China is actively promoting the widespread application of clean energy[1]. Wind power generation, solar power generation, and tidal power generation, among other sustainable energy forms, play an important role in this green transformation. Especially wind power generation, with its unique advantages, has been rapidly promoted and applied nationwide. However, wind power generation also has certain challenges. Due to its high dependence on natural wind, the amount of wind power generation is directly affected by weather conditions[2]. The intensity, direction, and stability of the wind all affect the power generation efficiency of the wind turbines[3-4]. The variability of weather factors makes the amount of wind power generation have a large degree of

randomness and uncertainty, bringing a significant impact on the stable power supply of the power grid[5]. The power grid is a highly complex system that needs to maintain a balance between power supply and demand. The instability of wind power generation may break this balance, leading to issues such as power grid voltage fluctuations and frequency instability[6]. This not only affects the normal operation of power equipment but may also cause local or even large-scale power outages[7]. Therefore, accurately forecasting wind power generation is particularly important. Through accurate forecasting, power dispatch departments can make adjustments in advance and reasonably arrange the operation of other power sources to make up for the fluctuations in wind power generation[8-9]. This can not only ensure the safe and stable operation of the power grid but also effectively improve the utilization rate of wind energy resources and reduce energy waste[10-11]. To achieve this goal, considering the situation of China's power system, the historical data of wind power generation is relatively accurate and complete[12]. This paper selects the expert system method and RBF neural network model to construct an optimized composite forecasting model, thereby improving forecasting accuracy[13]. The case study results show that the forecast is good in the prediction of wind power generation[14].

II. RBF NEURAL NETWORK FORECASTING MODEL

BP neural networks are an important branch in the field of machine learning and are widely used in pattern recognition, function approximation, data compression, and other areas. However, the learning process of BP networks, as a typical nonlinear optimization process, often faces the challenge of local minima. Local minima refer to the situation where the optimization algorithm may become trapped in a local optimal solution rather than a global optimal solution during the training

process. This can lead to slow convergence or even failure to converge, thereby affecting the learning effect.

To overcome the shortcomings of BP networks in solving the local minima problem, researchers are constantly exploring new neural network structures and algorithms. Among them, the Radial Basis Function (RBF) network has become an effective alternative due to its unique structure and performance advantages. RBF networks show higher efficiency in approximating nonlinear functions, and their convergence speed is also significantly faster than that of BP networks.

The RBF neural network is a multi-input single-output feedforward network structure. Its basic principle is to map the input space to a new space and perform linear combinations in that space to achieve approximation of nonlinear functions. As shown in Figure 1, the hidden layer of the RBF neural network is responsible for the nonlinear transformation of feature extraction, which remaps the input space to a completely new space. The output layer then completes the task of linear combination in this new space, thereby achieving the approximation of nonlinear functions.

In RBF neural networks, the adjustable parameters mainly include the weights of the linear combination and those parameters that control the shape of the basis functions. These parameters are adjusted during the training process to enable the network to better approximate the target function. During the training process, RBF networks typically use a method called "supervised learning," which trains the network based on known input-output sample pairs, allowing the network to learn the mapping relationship between input and output.

The hidden layer neurons of the RBF neural network usually use nonlinear functions as activation functions, among which the Gaussian function is most commonly used. The Gaussian function has good local approximation ability and smoothness, effectively capturing the local features of the input data. In the RBF network, each hidden layer neuron is associated with a specific center vector, which determines the response range and intensity of that neuron to the input data. At the same time, each hidden layer neuron also has a normalization parameter (also known as width or standard deviation) to control the shape and response range of the basis function.

In the RBF neural network, the nonlinear function adopted by the hidden layer neurons is usually a Gaussian function, as shown in formula (1):

$$G(\|X_{(i)}-T_i\|)=\exp\left[-\frac{1}{2}\left(\frac{\|X_{(i)}-T_i\|}{\delta_i}\right)^2\right] \quad (i=1,2,\dots,m) \quad (1)$$

where $X(t)=(x_{t1},x_{t2},\dots,x_{tn})$ is the t -th input sample in the training sample set; $T(i)=(t_1,t_2,\dots,t_n)$ is the center of the i -th hidden layer neuron; σ_i is the normalization parameter of the i -th hidden layer node.

During the training process, the center vectors and normalization parameters of the RBF network are typically initialized through clustering algorithms or random selection methods. Subsequently, through the iterative training process, the weights of the linear combination and the parameters of the basis functions are continuously adjusted to make the network's output better approximate the target output. Specifically,

optimization algorithms such as gradient descent or least squares can be used for the training process.

Due to the high efficiency and rapid convergence of the RBF network in approximating nonlinear functions, it has been widely applied to various practical problems, such as time series prediction, image recognition, and speech recognition. By reasonably designing the network structure and training algorithms, the RBF network can effectively model and predict complex nonlinear systems.

III. EXPERT SYSTEM CORRECTION METHOD

The essence of any knowledge base involved in an expert system is to simulate the thinking and decision-making process of an expert. The purpose of our research is to build a wind power generation correction system that can accurately simulate the professional knowledge, rich experience, and logical reasoning of senior dispatchers under situations such as sudden weather changes. Although the impact of climate change in

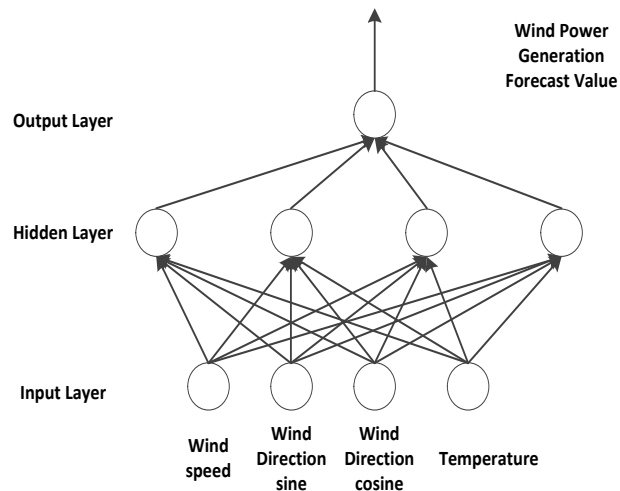


Figure 1. Principle of RBF Neural Network Forecasting Algorithm

different seasons on wind power generation varies, the correction process for each season is quite similar. Taking summer as an example, the correction process is explained. Regarding weather factors, we have set up a weather condition distinction coefficient as shown in Table 1.

Table 1 Weather Condition Distinction Coefficients

No.	Weather Condition	Distinction Coefficient
1	Heavy Snow	1
2	Moderate Snow	1.5
3	Light Snow	2
4	Thunderstorm	2.5
5	Heavy Rain	3
6	Moderate Rain	3.5
7	Light Rain	4
8	Overcast	4.5
9	Cloudy	5
10	Sunny	5.5

The calculation method for the wind power generation correction process is as follows: calculate the weather change coefficient according to formula (2):

$$d(k) = \alpha_1 \left(\frac{1}{2} (T_h(k) - T_h(k-1)) + \frac{1}{2} (T_l(k) - T_l(k-1)) \right) + \alpha_2 (l_{coc}(k) - l_{coc}(k-1)) \quad (2)$$

Where $d(k)$ is the weather change coefficient of the k -th day of the forecast; $T_h(k)$ is the highest temperature of the forecast day; $T_l(k)$ is the lowest temperature of the forecast day; $l_{coc}(k)$ is the weather condition distinction coefficient of the forecast day; α_1 is the temperature factor weight, taken as 0.5 in this paper; α_2 is the weather condition factor weight, also taken as 0.5 in this paper.

When facing sudden weather changes and special events, the expert system can quickly adjust the prediction strategy based on historical data and expert experience to ensure the accuracy of the prediction results.

Figure 2 illustrates the implementation process of the expert system method and the RBF neural network prediction model. First, by collecting historical data and weather information, we train the RBF neural network prediction model. Then, the prediction results of the RBF neural network are corrected using the expert system correction method. In this process, the expert system calculates the corrected prediction values based on the weather change coefficient and the prediction results of the RBF neural network. Finally, the corrected prediction values are output as the final wind power generation prediction results.

By introducing the expert system correction method, we have successfully integrated the professional knowledge and experience of senior dispatchers into the wind power generation prediction system. This not only improves the prediction accuracy but also enhances the adaptability and robustness of the prediction system.

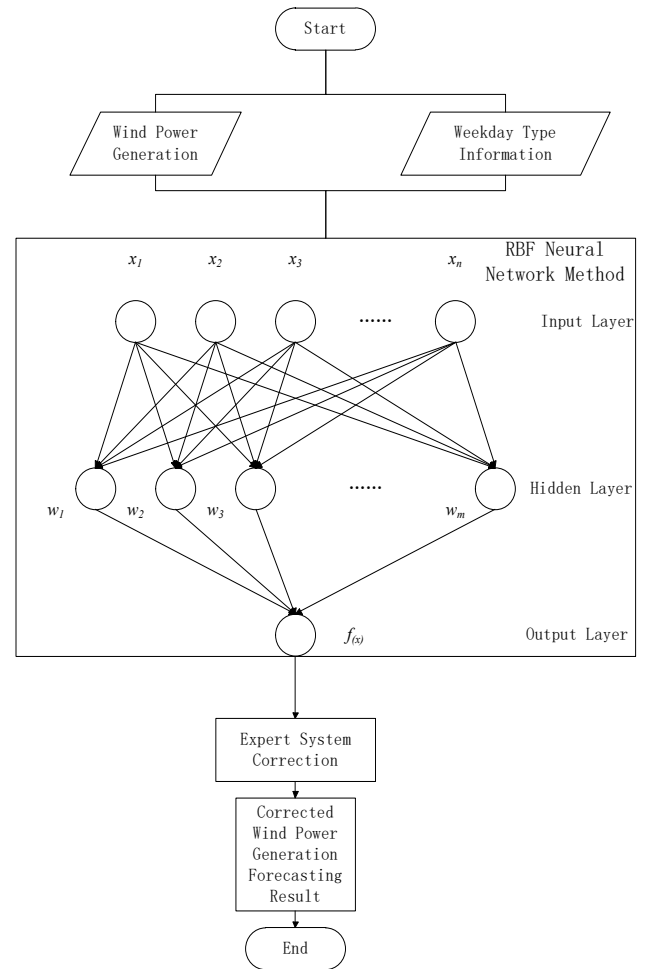


Figure 2. Implementation process of expert system method and RBF neural network forecasting model

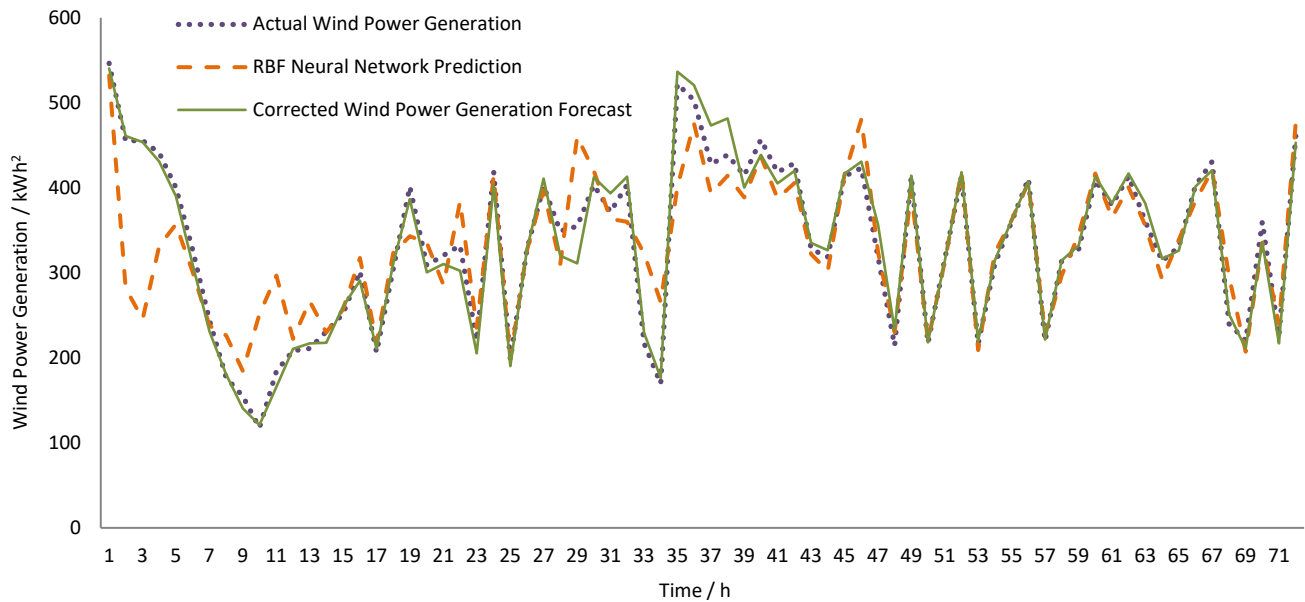


Figure 3. Comparison of predicted values and actual wind power generation of the two forecasting models

IV. CASE STUDY VERIFICATION

On the MATLAB platform, we conducted an in-depth case analysis of the wind power generation prediction problem for a wind farm in a certain province in H Province. First, we collected historical data from the wind farm, mainly including wind speed, wind direction, temperature, and the corresponding wind power generation at each time point. These data are the foundation for our predictive analysis.

To verify the accuracy of the prediction model, we selected a 72-hour period as the target for prediction. During this period, factors such as wind speed, wind direction, and temperature may undergo significant changes, thus posing high requirements for the stability and generalization ability of the prediction model.

We used two predictive models for comparison: a single RBF neural network prediction model and an RBF neural network prediction model combined with expert system methods. The single RBF neural network model learns the relationship between the characteristics of wind speed, wind direction, and temperature in the historical data and the wind power generation, attempting to find the underlying rules and make predictions. The RBF neural network prediction model combined with expert system methods, on the other hand, introduces expert experience on this basis, and refines the prediction results of the RBF neural network through the expert system to improve the accuracy of the prediction.

We implemented both predictive models through MATLAB programming and compared their prediction results. As shown in Figures 3 and 4, we can clearly see the comparison of wind power generation predictions by the two models during the 72-hour prediction period. The RBF neural network prediction model combined with the expert system method can more accurately predict wind power generation in most cases, demonstrating higher prediction accuracy and stability. This result verifies the effectiveness and superiority of the prediction method proposed in this paper.

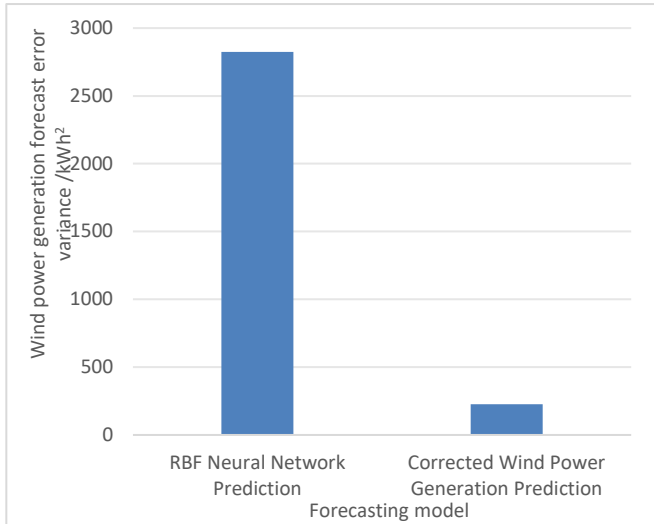


Figure 4 Comparison of prediction errors of the two forecasting models

V. CONCLUSIONS

This paper innovatively combines the expert system method with the RBF neural network forecasting model to accurately predict wind power generation. By fully integrating the logical reasoning ability of the expert system and the strong learning capability of the RBF neural network, we have constructed an efficient and accurate forecasting system. The prediction results are significant, and the system can closely track the changes in actual wind power generation, not only accurately predicting the trend but also greatly improving the prediction accuracy. This achievement is of great significance to the wind power generation industry, as it can help operators better plan power production, optimize resource allocation, reduce energy waste, and enhance the stability of wind power grid integration.

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